Ships In Harbor

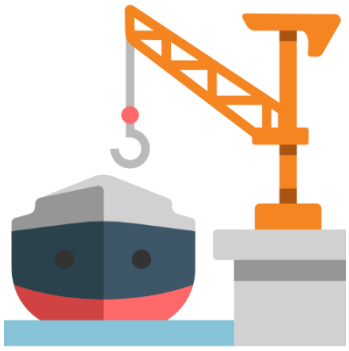
Programming L+Pr. (IP-18fPROGEG) – 2023/2024 spring semester

Your submitted solution is only **valid if it contains the following paragraphs** at the top of the program. Please **fill** in **your name** and **Neptun code**.

```
/*
<NAME>, <NEPTUN>
This solution was prepared and submitted by the student stated above for the
assignment of the Programming course. I declare that this solution is my own
work. I have not copied or used third party solutions. I have not passed my so-
lution to my classmates, neither made it public.
Students' regulation of Eötvös Loránd University (ELTE Regulations Vol. II.
74/C.§) states that as long as a student presents another student's work - or
at least the significant part of it - as his/her own performance, it will count
as a disciplinary fault. The most serious consequence of a disciplinary fault
can be dismissal of the student from the University.
*/
```

When submitting on TMS upload the program with the output enabled that only prints the result of the specific subtask e.g.: *if you are solving a) do NOT include the outputs for tasks b) c) d)***!**

Ships In Harbor



Ships arrive at a sea port, we know their arrival times (the number of the day) in chronological order. Arrivals were monitored for M days.

Create a program that gives the:

a) the number of days when no ship arrived;
b) a day (between day 2 and day M-1) on which a ship arrived and no ship arrived on the previous day and on the following days;
c) the length of the longest period during M days when no ship arrived;
d) the number of the day when the most ships arrived (in the case of several solutions, give the first);

Input

The first line of the *standard input* contains the number of ships ($1 \le N \le 1000$) and the number of days ($1 \le M \le 1000$). The next N line contains the day of each ship's arrival ($1 \le E_i \le M$).

Output

On the *standard output* give the result of each task separately as shown in the example.

1. subtask: The number of days when no ship arrived must be entered in the first line!

2. subtask: In the first line, you must write the number of a day on which a ship arrived, but not on the previous and the following day! In the case of several solutions, the number of the very first such day must be written; if there is no such day, then -1!

3. subtask: In the first line, the number of days of the longest period when no ships arrived must be written (0 if there were none)!

4. subtask: In the first line, write the number of a day when the most ships arrived (in the case of several solutions, the first)!

Example

Input	Output
7 45 3 6 6 42 42 42 43	41 3 35 42

Evaluation based on 10 input tester files:

Points: 10*(2+2+3+3)=10*10=100 points